AWS Cloud & DevOps Fundamentals

A simple, practical refresher for understanding tools instead of memorizing service names

How to use this guide
Whenever a concept feels unfamiliar, ask: What problem does it solve? How does it work? How does data flow? How is it secured? What usually breaks?

Worked example: The NextWork Build a Security Monitoring System project: AWS records access to a secret, detects the event, and sends an email alert.

Learning goal: Understand the job of each component, how components connect, and where to look when the system does not work.

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1. The five-question framework

Do not try to memorize every AWS service. Learn to explain each tool using these five questions. Use the same method for databases, networking, containers, CI/CD, monitoring, and cloud infrastructure.

Question	How to answer it
1. What problem does it solve?	Name the pain it removes. Example: Secrets Manager prevents passwords from being hardcoded in application code.
2. How does it work?	Know the basic parts and what each part does. Example: a metric filter searches logs and turns matching events into a metric.
3. How does data or traffic flow?	Trace the journey from start to finish: secret access → CloudTrail event → CloudWatch Logs → metric → alarm → SNS email.
4. How is it secured?	Check identity permissions, network access, encryption, secret handling, and least privilege.
5. What breaks, and how do I troubleshoot it?	Check permissions, configuration, connectivity, logs, resource limits, and whether each step passed data to the next.

Practice rule: If you can explain these five points without reading a definition, you understand the tool well enough to start using it.

2. The monitoring system at a glance

The project monitors access to a test secret. Each AWS service has one main job; together, they create a security-alert pipeline.

Step	Component	Job in plain English
1	AWS Secrets Manager	Stores the sensitive value safely.
2	AWS CloudTrail	Records the API activity when the secret is accessed.
3	Amazon S3	Keeps a durable copy of CloudTrail log files.
4	CloudWatch Logs	Receives logs so they can be searched and filtered.
5	Metric filter	Looks for GetSecretValue and records a metric.
6	CloudWatch alarm	Checks whether the metric meets the alert condition.
7	Amazon SNS	Sends email to subscribers when notified by the alarm.
Across the flow	AWS IAM	Controls which users and services may perform actions.

Remember: CloudTrail records activity. CloudWatch helps analyze and react to it. SNS delivers the notification. IAM permissions allow the components to do their jobs.

3. Core AWS concepts explained simply

3.1 AWS Secrets Manager

What it is: A secure place to store passwords, API keys, tokens, and other sensitive values.
Purpose: Keep secrets out of source code and insecure files.
Why use it: Applications can retrieve credentials when needed without hardcoding them.
Example: A web app retrieves its database password from Secrets Manager instead of storing it in Git.
Troubleshoot: Check secret name, Region, and permission to retrieve the value.

3.2 Key-value pairs

What it is: A label associated with the value it describes.
Purpose: Make secret data structured and understandable.
Example: username = app_user; password = a protected password.
Troubleshoot: Check that the expected key exists and the application parses the format.

3.3 Encryption and AWS KMS

What it is: Encryption scrambles information; AWS KMS helps manage encryption keys.
Purpose: Protect data confidentiality.
Example: A secret is stored encrypted and returned to an authorized caller.
Troubleshoot: Check encryption-key configuration and permission to use the key.

3.4 AWS CloudTrail

What it is: An activity recorder for AWS API actions, including who did what and when.
Purpose: Auditing, investigations, security monitoring, and troubleshooting.
Example: Retrieving TopSecretInfo creates a GetSecretValue event.
Troubleshoot: Check Region, time range, event source secretsmanager.amazonaws.com, and event name GetSecretValue.

3.5 Trail and Amazon S3

What it is: A trail configures delivery of selected CloudTrail events; S3 stores delivered log files.
Purpose: Keep logs for later investigations and retention.
Example: secrets-manager-trail delivers files to an S3 bucket.
Troubleshoot: Check trail logging, destination bucket, delivery delay, and bucket permissions.

3.6 CloudTrail event types and API activity

What it is: Management events record control-plane actions; data events cover certain operations on resources; Insights events identify unusual patterns; network activity events cover supported network actions.
Purpose: Select relevant activity to record.
Example: This project monitors GetSecretValue as a management event.
Troubleshoot: Check event selectors and read/write settings. Classifications depend on the API and service.

3.7 CloudWatch Logs and log groups

What it is: CloudWatch Logs collects and organizes logs. A log group is a collection of log streams; a stream contains log events.
Purpose: Search, inspect, filter, and retain logs.
Example: CloudTrail sends events to nextwork-secretsmanager-loggroup.
Troubleshoot: If there are no streams or events, check trail integration, IAM role, Region, delivery delay, and selected log group.

3.8 Metric filters and metrics

What it is: A metric is a numeric measurement over time. A metric filter searches incoming logs for a pattern and emits metric data when it matches.
Purpose: Turn text logs into measurable signals for graphs and alarms.
Example: GetSecretValue matches and records 1; a configured default value of 0 represents periods without matches

when applicable.
Troubleshoot: Check filter pattern, log group, metric namespace/name, and new matching logs.

3.9 CloudWatch alarm

What it is: A rule that evaluates a metric against a condition and changes state when the condition is met.
Purpose: Alert or trigger an action without a person watching continuously.
Example: Threshold >= 1 means the alarm can enter ALARM when the metric reaches at least one within its evaluation settings.
Troubleshoot: Check metric, statistic, period, threshold, evaluation settings, state, and alarm actions.

3.10 Amazon SNS

What it is: A publish/subscribe messaging service. A topic is the message channel; a subscription defines the destination.
Purpose: Deliver notifications to email or other supported endpoints.
Example: The alarm publishes to SecurityAlarms, and SNS emails a confirmed subscriber.
Troubleshoot: Confirm the email subscription and check that the alarm uses the correct topic and Region.

3.11 AWS IAM

What it is: IAM controls who can do what. Policies define permissions; roles provide permissions that trusted users or services can assume.
Purpose: Authorize actions safely.
Example: CloudTrail needs a role with suitable permission to deliver logs to CloudWatch Logs.
Troubleshoot: Read the exact AccessDenied action; check identity policy, resource policy, role trust, permission boundary, and organization controls.

3.12 AWS CLI and CloudShell

What it is: AWS CLI controls services with commands; CloudShell is a browser-based shell with AWS CLI access.
Purpose: Repeat tasks, automate work, and test API operations.
Example: aws secretsmanager get-secret-value --secret-id "TopSecretInfo" --region ap-south-1
Troubleshoot: Check syntax, resource name, Region, active identity, and permissions. Treat output containing secrets as sensitive.

4. How data flows through the system

1. An identity requests the secret A user or application calls GetSecretValue. IAM checks whether the identity is allowed to retrieve it.
2. CloudTrail records the event The event captures details about the API action. Verify it first in CloudTrail Event history.
3. Logs are delivered The trail can deliver log files to S3 and send a copy to CloudWatch Logs for monitoring.
4. The metric filter matches The filter searches log events for GetSecretValue and emits a numeric metric when it finds a match.
5. CloudWatch evaluates the metric The alarm checks the metric against its threshold and evaluation period.
6. SNS sends the notification When the alarm action runs, it publishes a message to the SNS topic, which delivers it to confirmed subscribers.

Fast diagnosis: If CloudTrail has no event, investigate the request or event lookup. If the event exists but CloudWatch has no logs, investigate delivery. If logs exist but the metric does not change, inspect the filter. If the metric changes but no email arrives, inspect the alarm action and SNS subscription.

5. Security principles to remember

Least privilege

Give each user, role, or service only the permissions required for its job. Broad permissions may make setup easier but increase risk. In production, prefer narrowly scoped permissions and remove temporary broad access.

Identity-based vs resource-based policies

Identity policies attach to users, groups, or roles. Resource policies attach to supported resources. Access can still be denied by explicit denies, permission boundaries, role trust, or organization-level controls.

Encryption vs access control

Encryption protects confidentiality; access control decides who may request or use the data. A secret can be encrypted and still require correct IAM permissions to retrieve.

Logging vs monitoring vs alerting

Logging records what happened. Monitoring evaluates signals. Alerting notifies someone when a condition needs attention. A log existing does not automatically mean an email alert is configured.

6. Troubleshooting checklist

1. Source event — Can I find GetSecretValue in CloudTrail Event history? Check Region, event source, event name, and time.

2. Log delivery — Is CloudTrail configured to send logs to the correct CloudWatch log group? Does it contain recent streams/events?

3. Metric filter — Does the filter use the intended pattern and log group? Does a new matching event change the metric?

4. Alarm — Is the alarm using the correct namespace and metric? Are statistic, period, threshold, evaluation settings, and actions correct?

5. SNS — Does the alarm point to the correct topic? Is the email subscription confirmed? Check spam/junk too.

6. IAM permissions — Which exact action was denied, and which identity or role made the request? Fix the smallest relevant permission set.

7. Time and Region — Are resources in the intended Region? Wait a few minutes for delivery and evaluation.

8. Cost and cleanup — After a lab, remove unneeded resources and review S3 objects, log retention, secrets, alarms, and other remaining resources.

> **Good troubleshooting habit:** Change one thing at a time, repeat the test, and record the result. That makes it easier to identify the change that fixed the issue.

7. Quick revision sheet

Term	Remember it as
Secrets Manager	Stores secrets securely.
CloudTrail	Records AWS API activity.
S3	Stores durable copies of log files.
CloudWatch Logs	Collects and searches logs.
Metric filter	Finds log patterns and emits metric data.
CloudWatch metric	A numeric measurement over time.
CloudWatch alarm	Checks a metric condition and changes state.
SNS	Delivers notifications to subscribers.
IAM	Controls who can perform which actions.
Least privilege	Grant only the permissions required.
CloudShell / CLI	Run AWS commands from a terminal.
Troubleshooting	Trace the pipeline from source event to notification.

> **Final mental model:** Store safely → record activity → collect logs → detect a pattern → evaluate a condition → notify the right person. Use IAM to control every step, and troubleshoot from the first failed stage.

This guide is a learning aid based on the supplied NextWork project text. AWS console labels and service behavior can change; check current AWS documentation for production implementations.